

Student ID 1 & 2: (Do this in pairs)

We use the following wildcard rules throughout the worksheet:

- `?` matches exactly one character.
- `*` matches any sequence of characters, including the empty sequence.
- A match must cover the *entire* string.

1. Warm-up: yes or no?

For each pair (s, p) below, decide whether the wildcard pattern p matches the entire string s . Write **Yes** or **No** in the last column.

String s	Pattern p	Match?
report_final.pdf	report*.pdf	
photo.png	*.jpg	
lab07.txt	lab??.txt	
acdcba	a*c?b	

2. Fill a DP table

Use the example $s = \text{acdcba}$ and $p = \text{a*c?b}$. Fill each cell with T or F.

	ϵ	a	*	?	b
ϵ					
a					
d					
c					
b					

What final answer do you read from the table?

3. Choosing the DP state

Suppose we want a dynamic programming solution. Let $\text{WM}[i][j]$ represent a statement about prefixes of the string and the pattern.

- (a) Complete the definition:

$WM[i][j] =$ _____

- (b) What should $WM[0][0]$ be, and why?

- (c) When can $WM[0][j]$ be true for $j > 0$?

4. Derive the recurrence

Using your table and the wildcard rules, complete the recurrence for $WM[i][j]$.

- (a) If $p[j - 1]$ is a normal character, what should the transition be?
- (b) If $p[j - 1]$ is `?`, what should the transition be?
- (c) If $p[j - 1]$ is `*`, explain the two choices that `*` gives you and write the transition.
- (d) Write down a recurrence that combines all cases and computes $WM[i][j]$ recursively from smaller subproblems.
- (e) What is the overall time complexity of the DP algorithm for a string of length m and a pattern of length n ?